

CYBER SECURITY

INTRODUCTION TO CRYPTOGRAPHY

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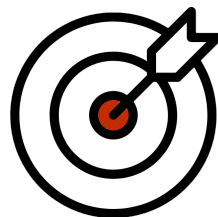
- Presentation goal;
- Definition;
- History;
- Main types;
- Logical Operations;

- Concept;
- Main Algorithms;

- Concept;
- Type of algorithms;
- ECB;
- CBC;
- Stream Cipher concept;
- Keystream;
- Main Algorithms;

- Overall Concept;
- Main Algorithms;

- Take Away;



TO LEARN ABOUT CRYPTOGRAPHY

1

**LEARN ABOUT CRYPTOGRAPHY
CORE CONCEPTS**

2

LEARN ITS MAIN ALGORITHMS

3

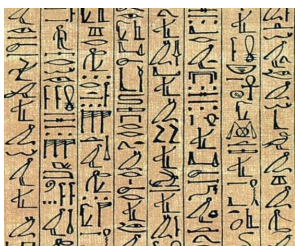
LEARN ABOUT ITS MAIN USAGE

DEFINITION

CRYPTOGRAPHY is the science related to **hiding information**. With the aim to provide a combination or all of the following: **Confidentiality, Integrity, Authentication** and **Non-repudiation**. This can be achieved through the design of **cipher algorithms**.

KRYPTOS (Greek) meaning HIDDEN, SECRET

NON STANDARD
HIEROGLYPHS



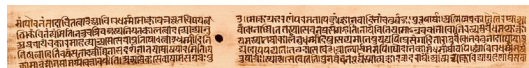
MESOPOTAMIA
CLAY TABLETS



SCYTALE
SPARTAN



KAMA SUTRA



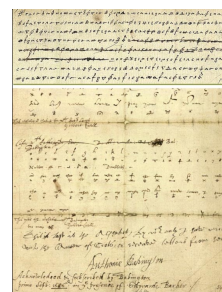
CAESAR SHIFT CIPHER



VIGENÈRE CIPHER



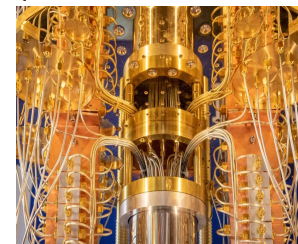
STEGANOGRAPHY
QUEENS OF SCOTS



WWII - ENIGMA



QUANTUM COMPUTER



1900 BC

1500 BC

700 BC

300 BC

50 BC

1553

1586

1940

2000s

MAIN TYPES

ENCRYPTION

HIDE DATA

DECRYPTION

REVEAL ORIGINAL DATA

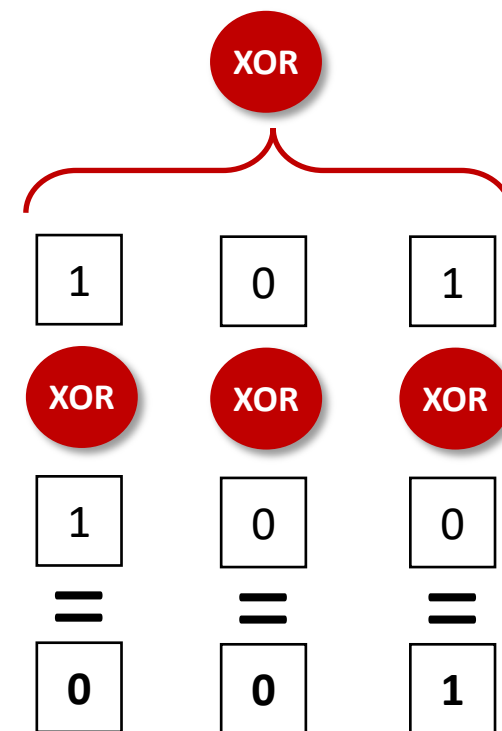
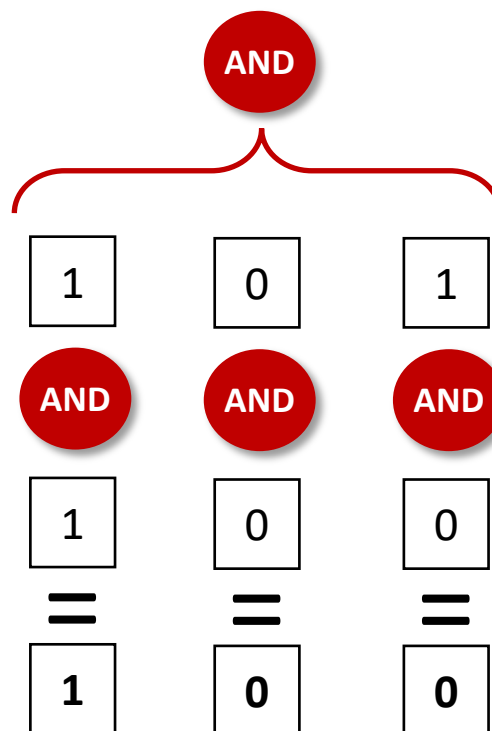
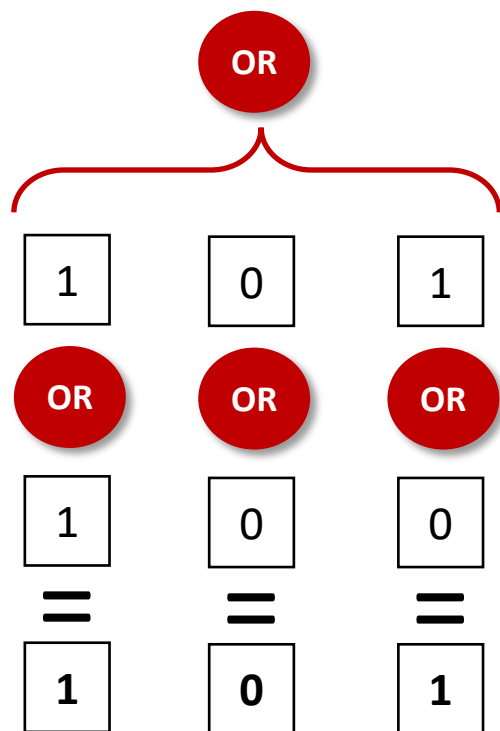
HASH

CREATES A UNIQUE SIGNATURE FROM DATA

STEGANOGRAPHY

HIDE DATA WITHIN OTHER DATA

LOGICAL OPERATIONS



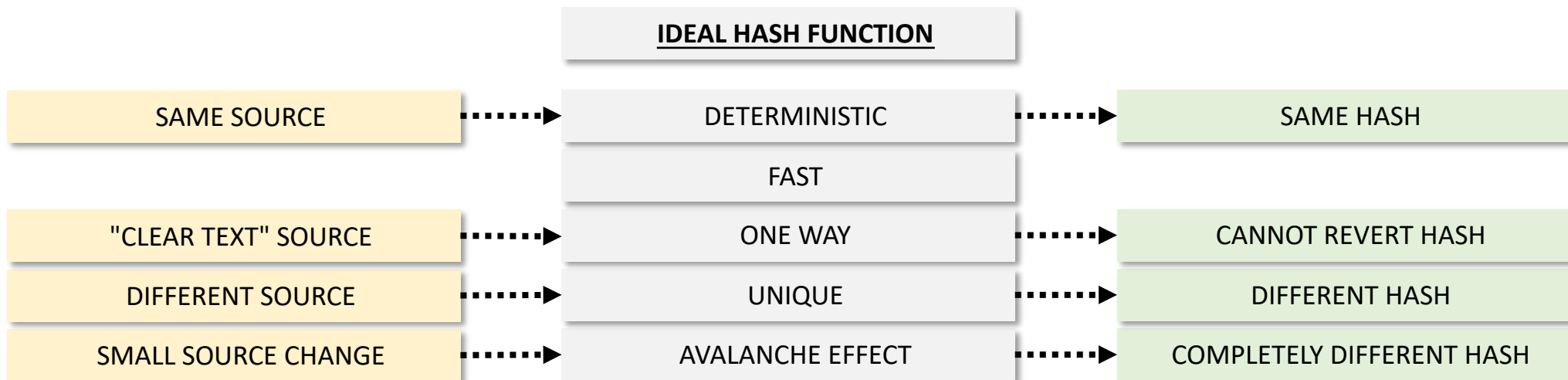
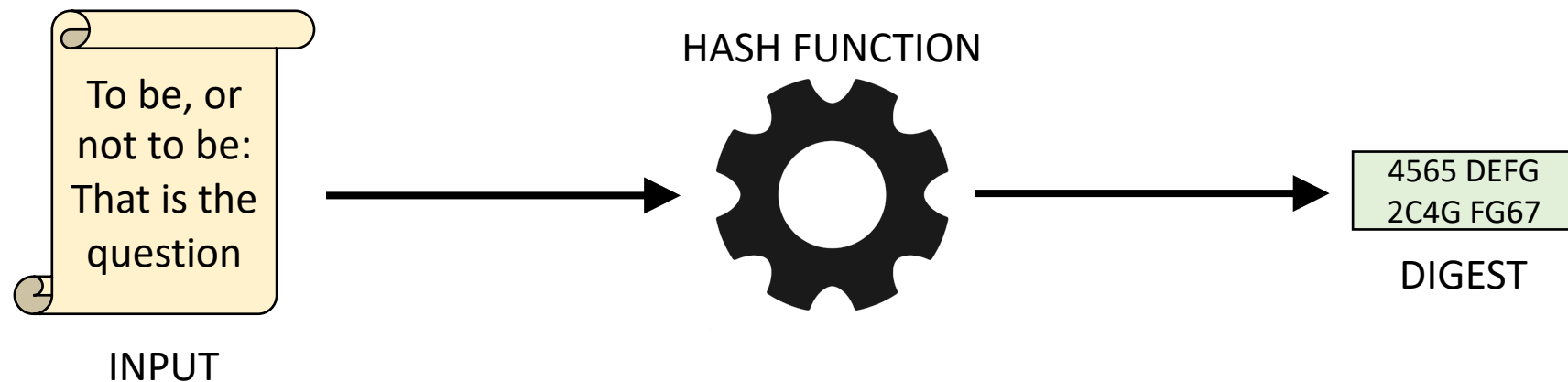
MORE LOGICAL OPERATIONS:

NOR

NAND

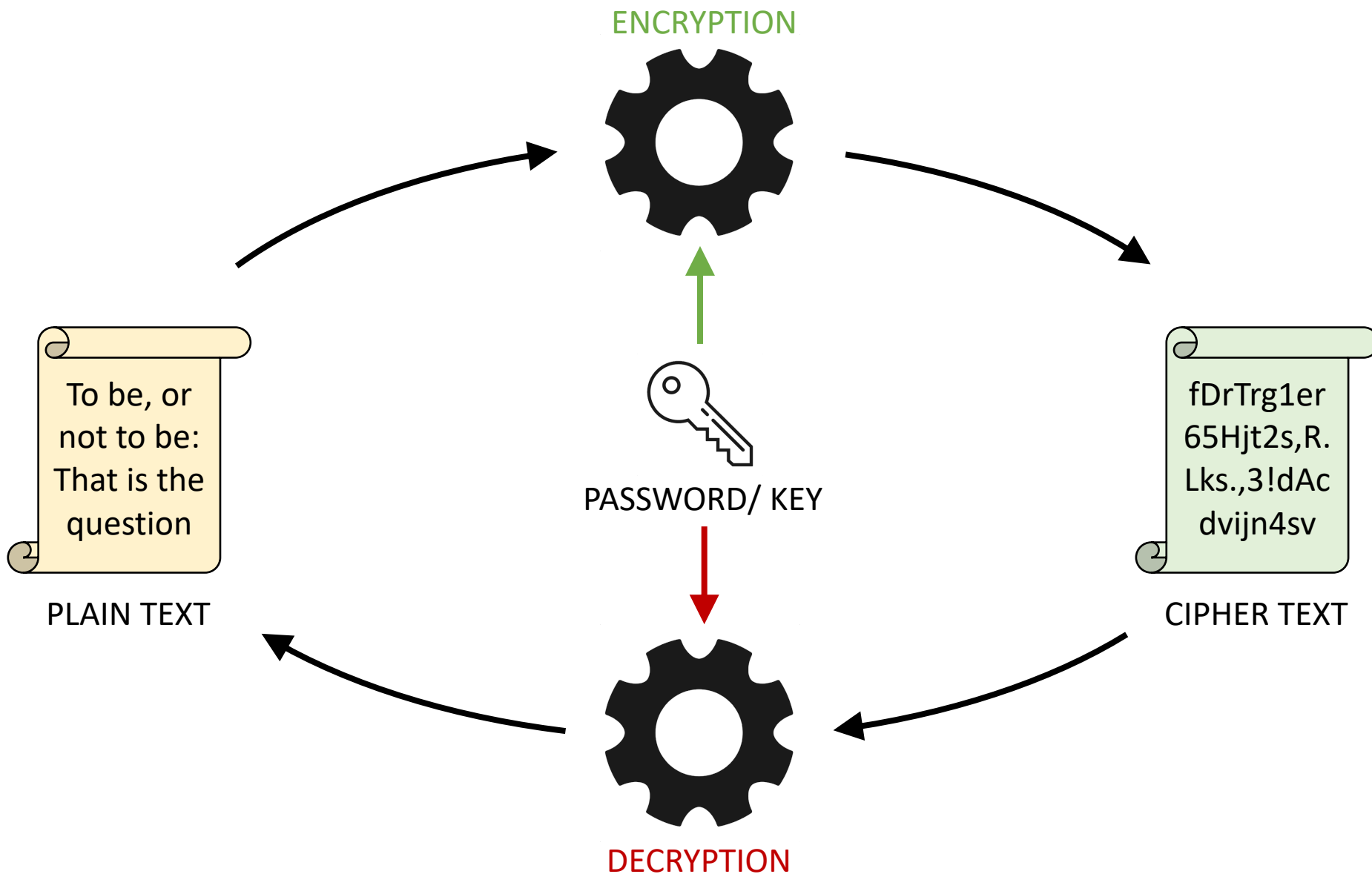
XNOR

NOT



MAIN ALGORITHMS

ALGORITHM	ORIGIN	SPECIFICITY	MAIN USAGE	RISKS
MD5	RONALD RIVEST 1991	128-BIT HASH VALUE 4 ROUNDS OF 16 OPERATIONS (XOR, AND, OR, NOT)	DATA INTEGRITY & CHECKSUM SSL DIGITAL CERT PASSWORDS	COLLISION ATTACK NO LONGER SUITABLE STILL WIDELY IN USE
SHA-1	NSA 1995	160-BIT HASH VALUE HEXADECIMAL 40 DIGITS LONG 80 ROUNDS	CHECKSUM, PWD, SSL/TLS CERT (NOT NOW) U.S GOV APPROVED	COLLISION ATTACK SHATTERED CHOSEN-PREFIX NO EASY/CHEAP ATTACKS
SHA-2	NSA 2001	224, 256, 384, 512-BIT HASH VALUE 64 OR 80 ROUNDS	CHECKSUM, PWD, BITCOIN, TLS, SSL, PGP, SSH, S/MIME, DKIM, DNSSEC, IPSEC	PARTIAL COLLISION ATTACK ONLY
SHA-3	NSA 2015	KECCACK FAMILY SPONGE CONSTRUCTION ABSORBED/SQUEEZED PADDING, PERMUTATION	CHECKSUM, PWD, ETHEREUM, TLS, SSL, DNSSEC, IPSEC	NONE KNOWN YET



TYPE OF CIPHER ALGORITHMS

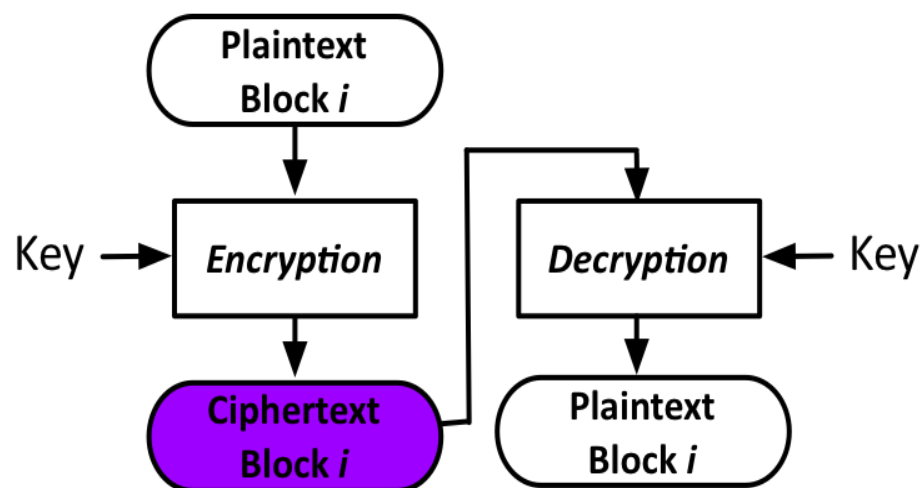
BLOCK CIPHER

**INPUT IS BROKEN INTO BLOCK OF DATA
EACH BLOCK OF DATA IS ENCRYPTED/DECRYPTED**

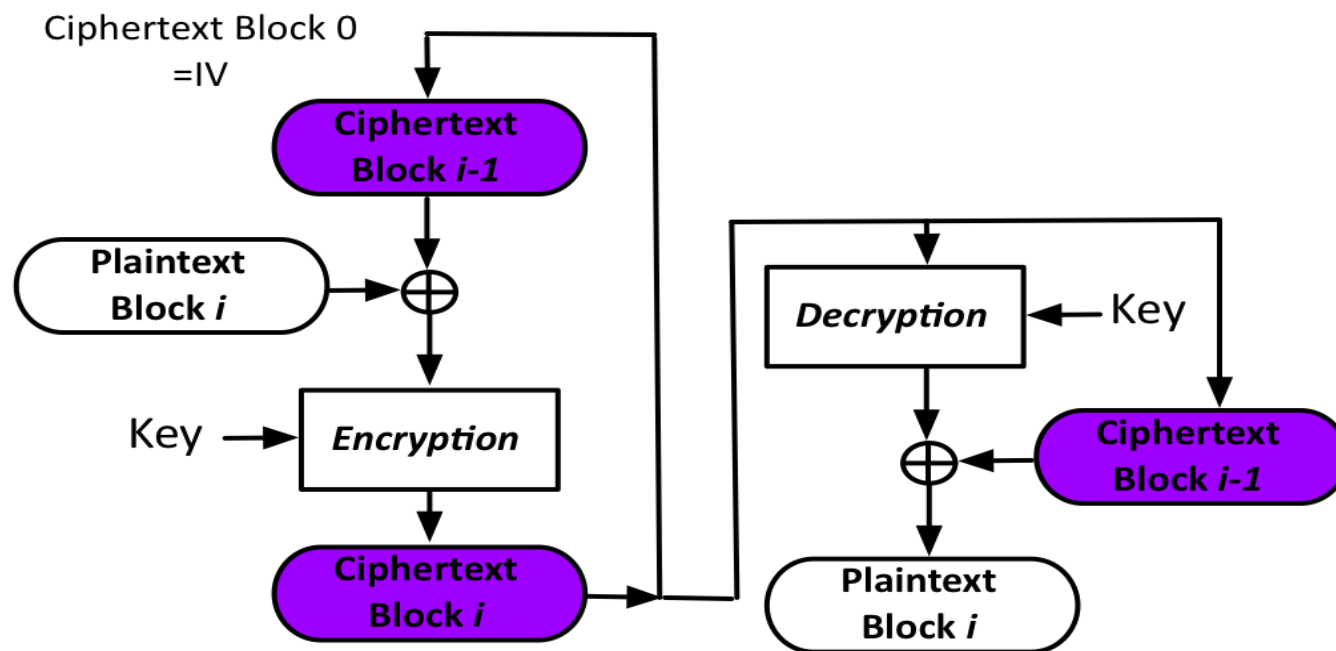
STREAM CIPHER

**INPUT IS BROKEN INTO A STREAM OF DATA
EACH BITS OF DATA IS ENCRYPTED/DECRYPTED**

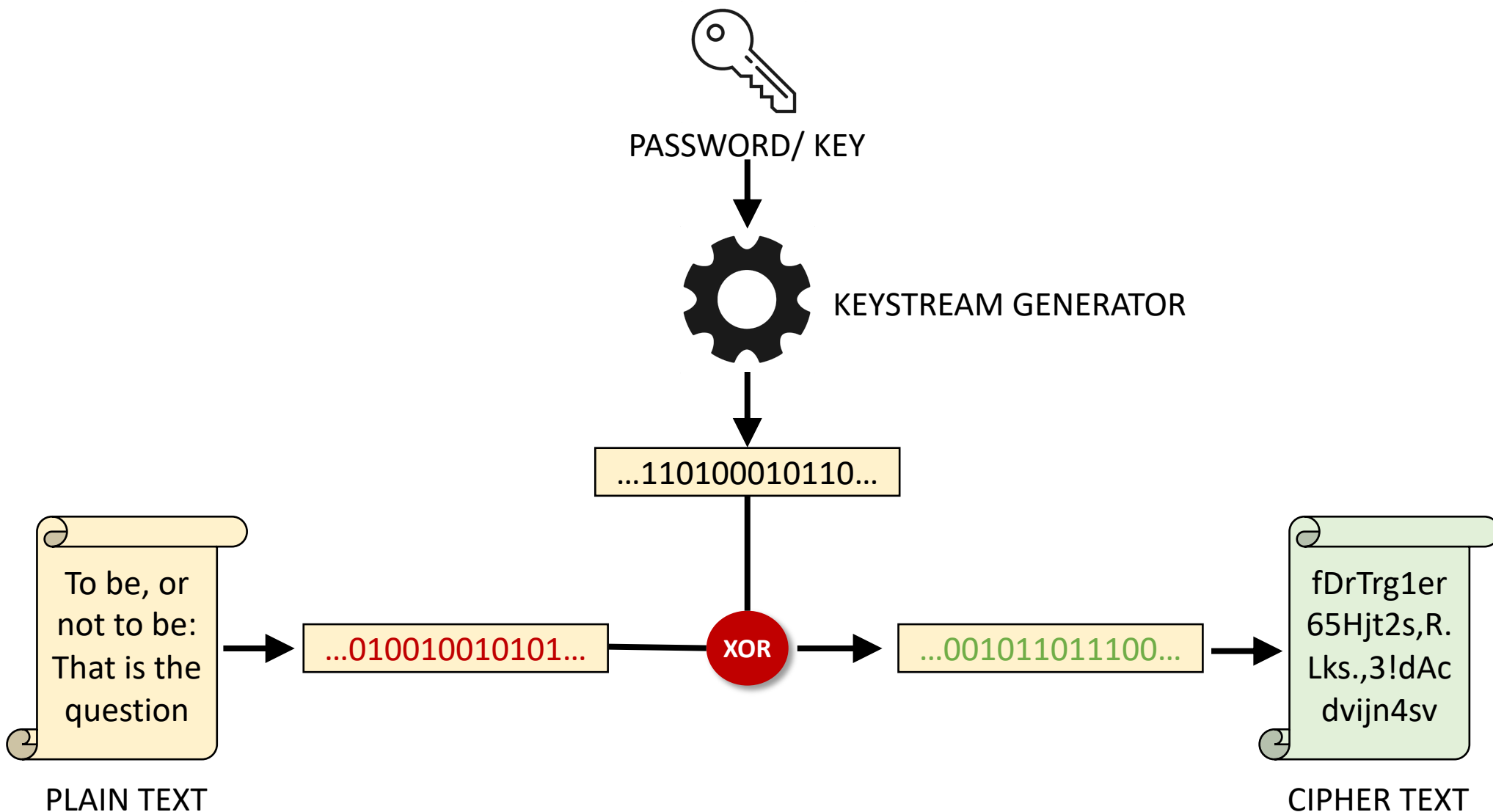
ELECTRONIC CODEBOOK MODE (ECB)



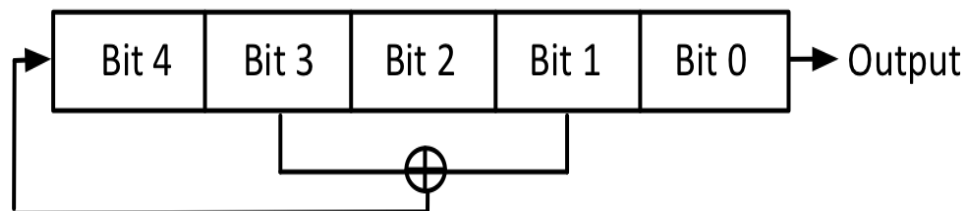
CHAIN BLOCK CHAINING MODE (CBC)



STREAM CIPHER CONCEPT



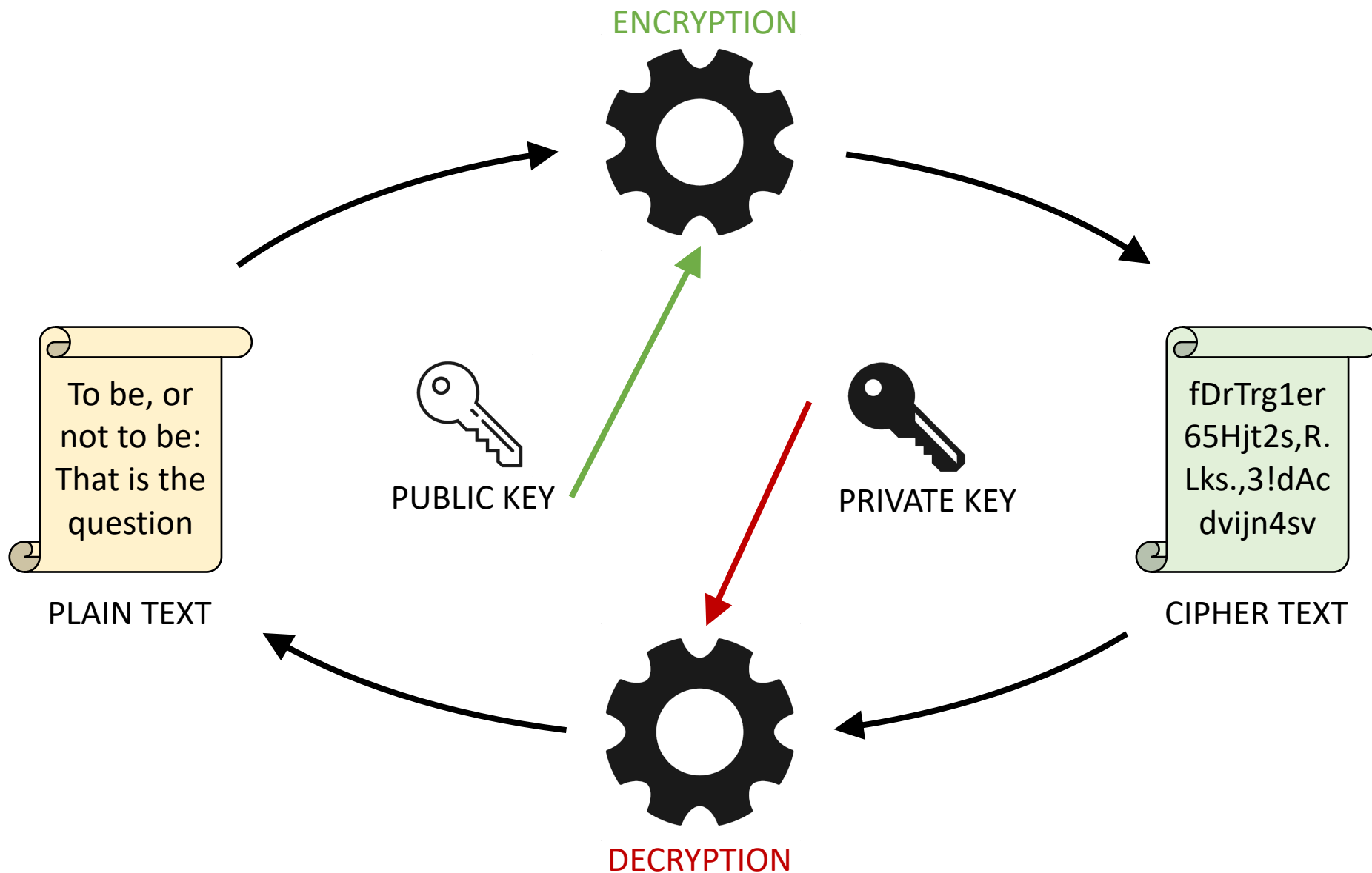
LINEAR FEEDBACK SHIFT REGISTER (LFSR)



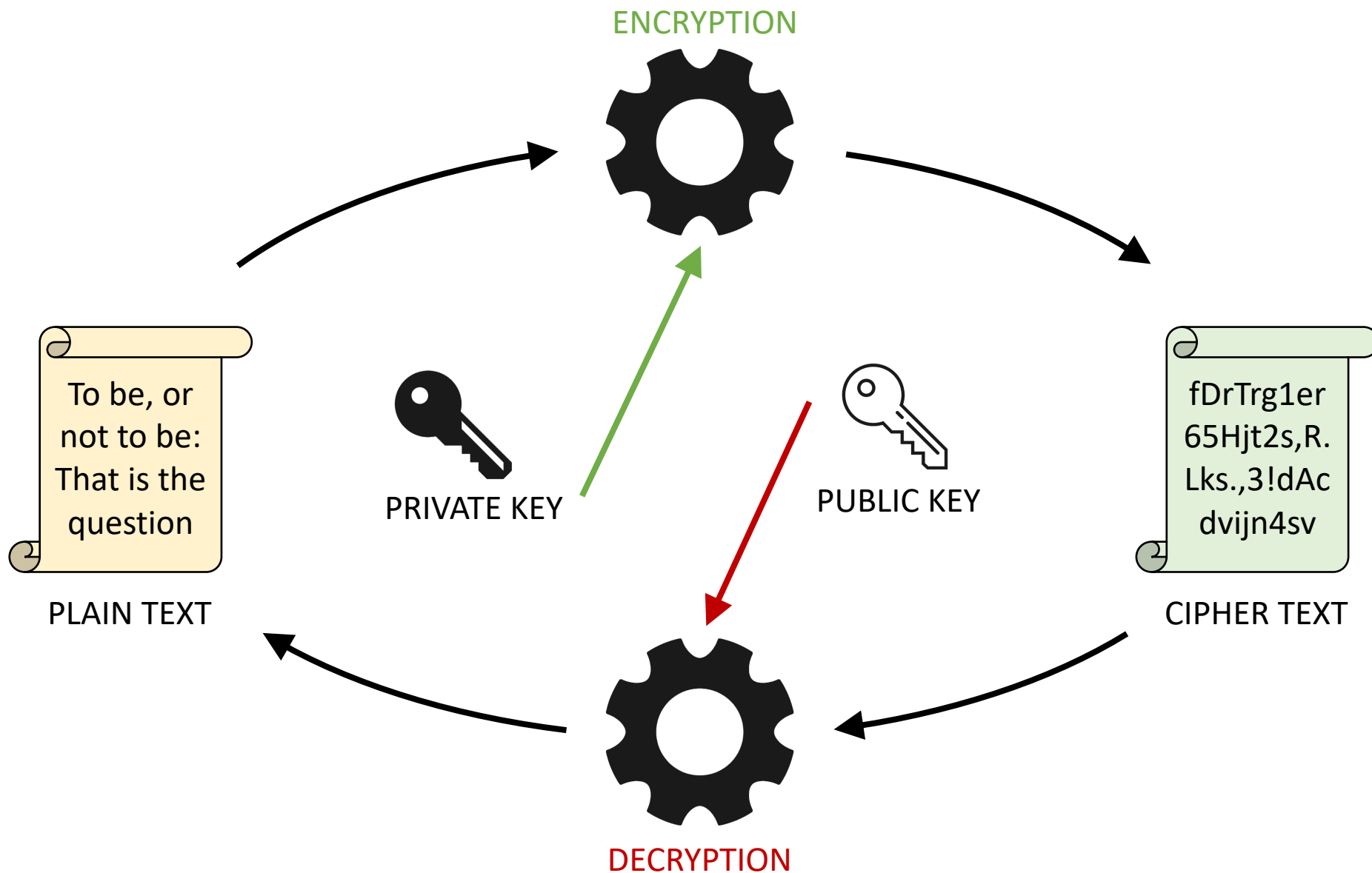
MAIN ALGORITHMS

ALGORITHM	ORIGIN	SPECIFICITY	MAIN USAGE	RISKS
RC4	RONALD RIVEST 1987	40-2018 BITS 1 ROUND PERMUTATION "S" KEY-SCHEDULING IV	WEP, WPA, BITTORRENT, MS REMOTE DESKTOP, KERBEROS, SKYPE, SSH	MULTI SESSION ATTACK ROYAL HOLLOWAY NO LONGER IN TLS
[DES] TRIPLE DES	IBM [1975], 1998	KEY: 168, 112, [56] BITS BLOCK: 64 BITS [16], 48-DES ROUNDS	STANDARD USED GLOBALLY (CREDIT CARD, PGP, ETC).	[S-BOX RESISTANT 20 YEARS EARLY] NO LONGER SECURED
TWOFISH	BRUCE SCHNEIER 1998	KEY: 128, 192, 256 BITS BLOCK: 128 BITS 16 ROUNDS	AES FINALIST OPENPGP	DIFFERENTIAL ATTACK
AES (RIJNDAEL)	VINCENT RIJMEN JOAN DAEMEN 1998	KEY: 128, 192, 256 BITS BLOCK: 128 BITS 10, 12 or 14 ROUNDS	NEW STANDARD DES/TDES REPLACEMENT SINCE 2002	XSL ATTACK RELATED KEY ATTACK NO PRACTICAL ATTACKS

OVERALL CONCEPT – RECEIVING DATA



OVERALL CONCEPT - AUTHENTICATION



MAIN ALGORITHMS

ALGORITHM	ORIGIN	SPECIFICITY	MAIN USAGE	RISKS
RSA	RIVEST SHAMIR ADLEMAN 1977	FACTORISATION KEY: 1,024 – 4,096 BITS 1 ROUND	TLS, SSH, ETC.	FACTORING PROBLEM NO LONGER A PROBLEM
ECDSA	NIST 2005	ELLIPTIC CURVE $y^2 = x^3 + ax + b$	SIGNATURE, KEY AGREEMENT, PRG	SONY/FAILOVERFLOW JAVA FLAWS
DSA	NIST 1991	ELLIPTIC CURVE	KEY GENERATION KEY DISTRIBUTION SIGNING SIGNATURE VERIF	SECRECY AND RANDOMNESS OF K
DIFFIE-HELMAN	WHITFIELD DIFFIE MARTIN HELLMAN 1976	MULTIPLICATIVE GROUP OF INTEGERS MODULO P	KEY AGREEMENT PROTOCOL	MAN IN THE MIDDLE LOGJAM ATTACK

TAKE AWAY

1

ONLY USE STANDARD ALGORITHMS

2

BEWARE OF THE IMPLEMENTATION

3

KEEP YOUR PRIVATE KEYS SAFE

4

BEWARE OF EXTERNAL FACTORS

5

CHECK FIPS CERTIFICATION DETAILS

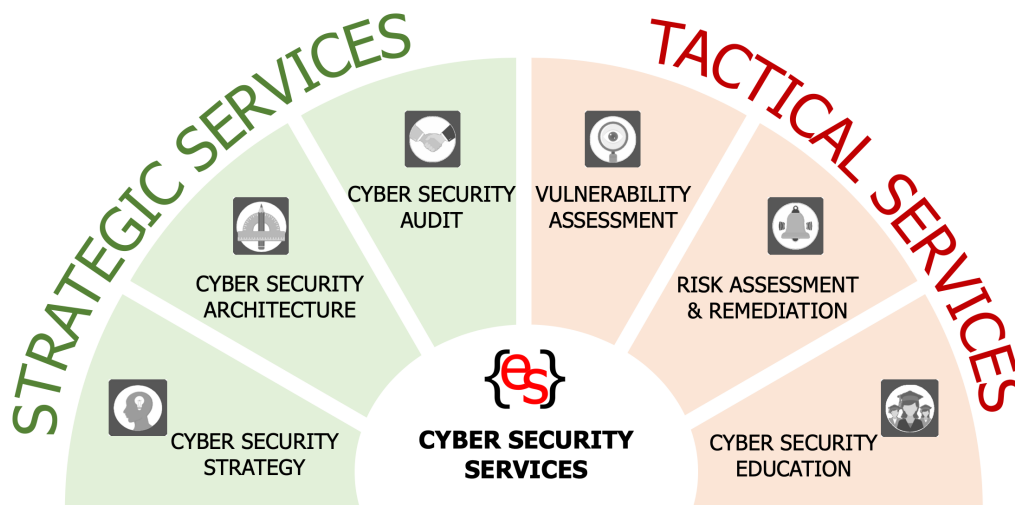
ABOUT ELYSIUMSECURITY LTD.

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